

GOVERNMENT OF INDIA - STATE RENEWABLE ENERGY DEVELOPMENT AGENCY

NOTICE INVITING TENDER: SREDA/SOLAR-SL/2026/04

TENDER FOR: Supply, Installation & 5-Year Maintenance of 90W Solar LED Street Lighting Systems

SECTION 3: TECHNICAL SPECIFICATIONS

1. SCOPE: Supply and installation of standalone solar street lighting systems for rural and urban roadways.

2. PHOTOVOLTAIC MODULE:

2.1 Crystalline silicon PV module with minimum peak power of 180 Wp at Standard Test Conditions (STC).

2.2 Module conversion efficiency shall be $\geq 18.5\%$ with fill factor $> 72\%$.

2.3 Mechanical load withstand: 2400 Pa wind load and 5400 Pa snow load.

2.4 Weatherproof terminal junction box with IP65 protection and bypass diodes.

2.5 PV module design qualification per IS 14286 / IEC 61215.

3. LED LUMINAIRE & OPTICS:

3.1 90W LED Luminaire with pressure die-cast aluminium LM6 housing.

3.2 Luminaire system efficacy ≥ 130 Lumens/Watt.

3.3 Correlated Color Temperature (CCT) 5700K (+/- 300K) and CRI ≥ 70 .

3.4 Optical compartment ingress protection rating IP65 / IP66.

3.5 Impact resistance rating IK08 toughened glass.

4. BATTERY & ENERGY STORAGE:

4.1 Lithium Ferro Phosphate (LiFePO₄) battery pack: 12.8V, 60Ah (768 Wh capacity).

4.2 Minimum 2500 cycles at 80% Depth of Discharge.

4.3 Integrated smart Battery Management System (BMS) with overcharge and cell balance.

4.4 Operating temperature: -10°C to 50°C.

5. CHARGE CONTROLLER:

5.1 MPPT charge controller with $\geq 98\%$ tracking efficiency and dusk-to-dawn sensor.

5.2 Built-in surge protection device withstanding 10 kV line-to-earth surges.

6. MAST & POLE:

6.1 7-meter octagonal hot-dip galvanized steel pole with zinc coating ≥ 400 g/m².

6.2 Structural withstand against gust wind speeds up to 150 km/h.

7. WARRANTY & COMPLIANCE:

7.1 5 years comprehensive on-site warranty for entire system.

7.2 Mandatory BIS CRS registration for battery and ISI Mark for luminaire.